

Sept 14, 2021

$$y_i = E[y_i | x_i] + e_i$$

$$E[e_i | x_i] = 0$$

$$E[y_i | x_i] \sim x_i' \beta$$

$$\begin{array}{cc} x_i & \beta \\ k \times 1 & k \times 1 \end{array}$$

$$y_i = \underset{1 \times k}{x_i'} \underset{k \times 1}{\beta} + e_i$$

semi parametric

$$Y = X \beta + e$$

$$\begin{array}{ccccc} n \times 1 & n \times k & k \times 1 & n \times 1 \end{array}$$

$$S(b) = \sum_{i=1}^n e_i^2(b) = \sum_{i=1}^n (y_i - x_i' b)^2$$

$$\text{F.O.C.} \quad \frac{dS(b)}{db} = -2 \sum_i x_i (y_i - x_i' b) \quad k \text{ eqs.}$$

$$\hat{\beta} \text{ is the } b \text{ s.t. } \left. \frac{dS(b)}{db} \right|_{\hat{\beta}} = 0$$

$$X' (Y - X \hat{\beta}) = 0$$

$$\sum x_i y_i = \left(\sum_i x_i x_i' \right) \hat{\beta}$$

$$\left(\sum x_i x_i' \right)^{-1} \sum x_i y_i = \hat{\beta}$$

$$\text{rank}(X) = k$$

$$(X'X)^{-1} \text{ exists}$$

$$\hat{\beta} = \overbrace{(X'X)^{-1}}^{k \times k} \underbrace{X'Y}_{n \times k}$$

$$= \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum x_i y_i$$
$$\begin{array}{cc} k \times k & 1 \times k \end{array}$$

orthogonality condition.

$$y = x\beta + e$$

$$e = y - x\beta$$

$$y_i = x_i'\beta + e_i$$

$$E[y_i | x_i] = x_i'\beta + \underbrace{E[e_i | x_i]}$$

$$\rightarrow E[e_i | x_i] = 0$$

$$E[e_i | x_i] = 0 \Rightarrow E[e_i x_i] = 0$$

$$E[e_i x_i] = 0$$

population orthogonality condition

$$E[(y_i - x_i'\beta) x_i] = 0$$

$$\frac{1}{n} \sum (y_i - x_i'\hat{\beta}) x_i' = 0$$

pop.

$$E[e_i x_i] = 0$$

$$x' e = 0$$

sample

$$\frac{1}{n} \sum \hat{e}_i x_i' = 0$$

$$x' \hat{e} = 0$$

OLS $\hat{\beta} = (x'x)^{-1} x'y$

$k=1$ $x = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$
 $n \times 1$

$$x'x = \begin{bmatrix} 1 & 1 & \dots & 1 \\ & & x' & \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = n$$

$$(x'x)^{-1} = \frac{1}{n}$$

$$x'y = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n y_i$$

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1} X'Y \\ &= \frac{1}{n} \sum y_i = \bar{y}\end{aligned}$$

Projection Matrices.

$$Y = X\beta + e$$

$$= X\hat{\beta} + \hat{e}$$

$$\hat{e} = Y - X\hat{\beta} = Y - \underbrace{X(X'X)^{-1}X'}_{P_X} Y$$

$$P_X = X(X'X)^{-1}X'$$

$$M_X = I - P_X$$

$$\begin{aligned}\hat{e} &= Y - P_X Y \\ &= (I - P_X) Y \\ &= M_X Y.\end{aligned}$$

$$\hat{y} = P_X Y$$

$$\begin{aligned}\hat{e} &= M_X Y \\ &= (I - P_X) Y.\end{aligned}$$

P_X and M_X are idempotent matrices

①. $P = PP$, $M = MM$

②. symmetric, $P = P'$ ($M = M'$)

③. eigenvalues of 0, 1

④. $\text{trace}(P) = K$ $\text{trace}(M) = n - K$.

Projections Theorem. $y = \hat{y} + \hat{e}$
 There exists $\hat{y}$ s.t. $\hat{y}' \hat{e} = 0$

$$K \text{ equations} \left| \begin{array}{c} x_1' \hat{e} \\ \vdots \\ x_K' \hat{e} \end{array} \right. \quad \left(\begin{array}{c} \sum x_{i1} \hat{e}_i \\ \vdots \\ \sum x_{iK} \hat{e}_i \end{array} \right) \quad \begin{array}{l} \text{normal} \\ \text{equations} \end{array}$$

$$x_{i1} \quad \dots \quad x_{iK}$$

$$x_i = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$\sum x_{i1} \hat{e}_i = 0$$

$$\text{when } x_{i1} = 1 \quad \forall i$$

$$\frac{1}{n} \sum \hat{e}_i = 0$$

① sample residuals are mean zero,

$$\text{②. } \overline{\hat{y}} = \bar{y}$$

Frisch - Waugh - Lovell Theorem.

$$y = x_1 \beta_1 + x_2 \beta_2 + e \quad \begin{array}{l} \text{Long} \\ \text{regression} \end{array}$$

Alternative approach.

①. Regress y on $x_1 \rightarrow$ residuals $\tilde{y} = M_1 y$

②. Regress x_2 on $x_1 \rightarrow$ residuals $\tilde{x}_2 = M_1 x_2$

③. reg. $\tilde{y}$ on $\tilde{x}_2 \rightarrow \tilde{\beta}_2$

$$\text{Theorem: } \hat{\beta}_2 = \tilde{\beta}_2$$

Example. $X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$y = \beta_1 + x_2 \beta_2 + e$$

$$(y - \bar{y}) = \beta_2 (x_2 - \bar{x}_2) + e$$

Special case when y on x_2 alone
will give same estimate as $\hat{\beta}_2$

$$\tilde{\beta} = (\tilde{x}_2' \tilde{x}_2)^{-1} \tilde{x}_2' \tilde{y}$$

$$\tilde{x}_2 = M_1 x_2$$

$$\tilde{x}_2' \tilde{y} = x_2' M_1 M_1 y = x_2' M_1 y$$

$$\tilde{x}_2 = M_1 x_2 = x_2 - x_1 (x_1' x_1)^{-1} x_1' x_2$$

$$\text{if } x_1 \perp x_2 = 0$$

$$\Rightarrow \tilde{x}_2 = x_2$$

$$\Rightarrow \tilde{\beta} = (x_2' x_2)^{-1} x_2' y$$

Goodness of Fit

$$y = \hat{y} + \hat{e}$$

$$y' y = (\hat{y} + \hat{e})' (\hat{y} + \hat{e})$$

$$= \hat{y}' \hat{y} + \hat{e}' \hat{e}$$

$\hat{e}' \hat{y}$
by projection
theorem.

$$R^2 = 1 - \frac{\hat{e}' \hat{e}}{y' y}$$

↓
centered R^2

$$M_1 Y = M_1 \hat{Y} + M_1 \hat{e}$$

$$Y' M_1 Y = \beta' X' M_1 X \beta + \hat{e}' \hat{e}$$

$M_1 = I - P$
 $x_i = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

$\downarrow$ $\downarrow$ $\downarrow$
 total SSR = explained + unexplained.

$$\text{centered } R^2 = \frac{\hat{\beta}' X' M_1 X \beta}{Y' M_1 Y} = 1 - \frac{\hat{e}' \hat{e}}{Y' M_1 Y}$$

$0 \leq R^2 \leq 1$ if a constant is included as regressor, non decreasing in k .

$$\text{adjusted } R^2 = 1 - \frac{\hat{e}' \hat{e} / (n-k)}{(Y - \bar{Y})'(Y - \bar{Y}) / (n-1)}$$

Is linear model restrictive?

$$(i) \quad y_i = A x_i^\beta \exp(e_i)$$

log linear model

$$\log y_i = \alpha + \beta \log x_i + e_i$$

$$(ii) \quad \text{sem-log model}$$

$$\log y_i = \beta x_i + e_i$$

iii) interaction effect.

$$m(x_1, x_2) = x_1 \beta_1 + x_2 \beta_2 + \underbrace{x_1 x_2}_{\text{interaction}} \beta_3 + \epsilon.$$

$$\text{iv), } D_1 = \begin{cases} 1 & \text{female} \\ 0 & \text{male} \end{cases} \quad D_2 = \begin{cases} 10 & \text{female} \\ 1 & \text{male} \end{cases}$$

$$\text{a, } w_i = \beta_0 + \beta_1 D_1 + \epsilon$$

$$\text{b, } w_i = \beta_0 + \beta_2 D_2 + \epsilon \text{ em.}$$

$$\text{c, } w = b_1 D_1 + b_2 D_2 + \epsilon \text{ em.}$$

↓
no constant.

Dummy Variable trap: we cannot include D_1, D_2 in same regression with a constant because $D_1 + D_2 = 1$.

multicollinearity.

$$\frac{1}{n} (X'X) = \begin{pmatrix} \hat{\text{var}}(x_1) & \hat{\text{cov}}(x_1, x_2) \\ \hat{\text{cov}}(x_2, x_1) & \hat{\text{var}}(x_2) \end{pmatrix}$$

$$\begin{pmatrix} \hat{\sigma}_1^2 & 0 \\ 0 & \hat{\sigma}_2^2 \end{pmatrix} \begin{pmatrix} 1 & \hat{\rho}_{12} \\ \hat{\rho}_{12} & 1 \end{pmatrix} \begin{pmatrix} \hat{\sigma}_1^2 & 0 \\ 0 & \hat{\sigma}_2^2 \end{pmatrix}$$

$$(X'X)^{-1} \approx (X'X + \lambda I)^{-1}$$